ERDŐS PROBLEM #848 HOLDS FOR EVERY N: A COMPUTER-ASSISTED COMPLETION OF SAWHNEY'S THEOREM

BEN ZANGHI

ABSTRACT. Erdős and Sárközy asked whether the largest $A \subseteq \{1, \dots, N\}$ such that $ab+1$ is never squarefree for $a, b \in A$ is given by the residue class 7 mod 25. Sawhney proved this for all sufficiently large N; the best explicit threshold previously available was $N_0 = \exp(1420)$. We prove the statement for every $N \geq 1$ (with $a = b$ allowed). The proof has three parts: an exact computation for $N \leq 10^4$, a dynamically maintained colouring certificate for $10^4 \leq N \leq 10^9$ that is replayed by an independent verifier, and a new elementary explicit argument for $N \geq 5 \cdot 10^8$ whose numerical inequalities are checked in interval arithmetic. The argument for large N is self-contained and does not use Sawhney's stability theorem.

Disclosure. This work was carried out with substantial assistance from an AI system (Claude, Anthropic), which wrote the code and drafted the argument and this manuscript. All computations are reproducible from the accompanying code; see Section 6. An independent AI review re-implemented the analytic bounds without access to our code. The manuscript has not yet been checked by a human expert, and the author welcomes scrutiny.

1. INTRODUCTION

For $N \geq 1$ call $A \subseteq [1, N]$ *admissible* if $ab+1$ is not squarefree for all $a, b \in A$ (including $a = b$). Let

$$M(N) = \#\{n \leq N : \ n \equiv 7 \pmod{25}\}.$$

The set $P_7(N) = \{n \leq N : n \equiv 7 \ (25)\}$ is admissible since $ab + 1 \equiv 50 \equiv 0 \pmod{25}$. Erdős and Sárközy [1, p. 239] asked whether it is maximal; this is Problem #848 in Bloom's database [2]. Van Doorn observed that every admissible set has density at most $2\sum_{p \equiv 1(4)} p^{-2} \approx 0.108$, Weisenberg improved this to ≈ 0.105, and Sawhney [3] proved that the answer is yes for all sufficiently large N, together with a stability statement. An explicit version of Sawhney's argument, produced with AI assistance by Sothanaphan and others, gives the threshold $N_0 = \exp(1420)$ [4]. Direct computation had been carried out for N up to a few hundred.

Theorem 1. *For every $N \geq 1$ and every admissible $A \subseteq [1, N]$ we have $|A| \leq M(N)$. In particular, the residue class 7 mod 25 is a maximum admissible set for every N.*

Remark 2. If one only requires $ab + 1$ to be non-squarefree for $a \neq b$, the statement is different for small N: Cambie noted that $\{19, 33, 35, 46, 53\}$ is such a set. Our argument uses the diagonal condition in an essential way, through the restriction that $a^2 + 1$ is non-squarefree, and does not address that variant.

Strategy. Let G_N be the graph whose vertices are the $n \leq N$ with $n^2 + 1$ non-squarefree, with $a \sim b$ (for $a \neq b$) iff $ab + 1$ is non-squarefree. Admissible sets are exactly the cliques of G_N, so Theorem 1 says $\omega(G_N) = M(N)$. We prove it in three overlapping ranges:

Date: Draft of 12 October 2026 — not yet refereed.

range	method	section
$1 \le N \le 10^4$	exact clique number (two independent programs)	3
$9982 \le N \le 10^9$	colouring certificate + independent verifier	4
$N \ge 5 \cdot 10^8$	explicit inequalities (interval arithmetic)	5

2. Preliminaries

Since $n^2 \equiv 0, 1 \pmod 4$, we have $4 \nmid n^2+1$. If $p \equiv 3 \pmod 4$ then $p \nmid n^2+1$. Hence n^2+1 is non-squarefree iff $q^2 \mid n^2+1$ for some prime $q \equiv 1 \pmod 4$. For such q let $S_q = \{n : q^2 \mid n^2+1\}$, which is the union of two residue classes $\pm i_q \bmod q^2$. Since $25 \mid n^2+1 \iff n \equiv \pm 7 \pmod{25}$, the vertex set of G_N is the disjoint union

$$P_7 = \{n \equiv 7\}, \qquad P_{18} = \{n \equiv 18\}, \qquad R \subseteq \bigcup_{q \ge 13,\ q \equiv 1 (4)} S_q,$$

where all three are intersected with $[1, N]$ and R denotes the remaining vertices. Note $|P_{18}| \le |P_7| = M(N)$, and both P_7 and P_{18} are cliques.

For a clique A we write $X = A \cap P_7$, $Y = A \cap P_{18}$ and $C = A \cap R$.

Lemma 3 (witness facts)**.** *Let w, z, h, r be positive integers.*

(i) If $z \equiv 7 \pmod{25}$, then $25 \mid wz+1$ iff $w \equiv 7 \pmod{25}$. If $z \equiv 18 \pmod{25}$, then $25 \mid wz+1$ iff $w \equiv 18 \pmod{25}$.

(ii) If $h \equiv \pm 7 \pmod{25}$ and $r \not\equiv \pm 7 \pmod{25}$, then $25 \nmid hr+1$.

(iii) $4 \mid wz+1$ iff $wz \equiv 3 \pmod 4$, i.e. iff $\{w \bmod 4, z \bmod 4\} = \{1, 3\}$.

Proof. (i) $7^{-1} \equiv 18$ and $18^{-1} \equiv 7 \pmod{25}$, so $7w \equiv -1$ iff $w \equiv -18 \equiv 7$, and similarly for 18. (ii) $h^2 \equiv -1$ gives $h^{-1} \equiv -h$, so $25 \mid hr+1$ iff $r \equiv -h^{-1} \equiv h \pmod{25}$, which is excluded. (iii) This is immediate. □

Consequently, if A is a clique, then every element of $Y \cup C$ is a *witness* for X: X lies in its neighbourhood, and for $f \in Y \cup C$ and $z \in X$, $fz+1$ must be divisible by p^2 for some prime $p \ne 5$. Likewise $X \cup C$ are witnesses for Y, and $X \cup Y$ are witnesses for C. Whether $p = 2$ is available is decided by residues mod 4.

3. Small N

For every $N \le 10^4$ we computed $\omega(G_N)$ exactly with a branch-and-bound clique algorithm using greedy colouring bounds on bitsets (`maxclique_small.c`). In all cases $\omega(G_N) = M(N)$. An independent implementation in Python (Bron–Kerbosch with pivoting and naive squarefree tests, `bk_check.py`) confirms this for all $N \le 5000$.

Remark 4. G_N is not always $M(N)$-colourable *monotonically*. For instance, at $N = 43$ the vertices are $7, 18, 32, 38, 41, 43$. Here $18 \sim 43$, $7 \sim 32$ and $43 \sim 32$; a 2-colouring exists ($\{7, 43, 38\}$, $\{32, 18, 41\}$), but it cannot be extended consistently from smaller N. This is why the certificate below starts at $N_0 = 9982$.

4. The colouring certificate for $9982 \le N \le 10^9$

If the vertex set of G_N is covered by $M(N)$ independent sets, then $\omega(G_N) \le M(N)$. We construct such covers for all N in the range at once.

Proposition 5. *Let $N_0 = 9982$ and $N_1 = 10^9$. There is a sequence of operations, recorded in a log file, of three types: create anchor x ($x \equiv 7 \bmod 25$); add v to class x; and move u to class x. It has the following properties.*

(a) *All anchors $x \le N_0$ are created first. Anchors $x > N_0$ are created in increasing order, interleaved with the adds.*
(b) *Every vertex $v \le N_1$ of G_{N_1} with $v \not\equiv 7 \pmod{25}$ is added exactly once. The adds occur in increasing order of v.*
(c) *Every add or move into class x happens at a time t, namely the vertex currently being inserted, with $x \le \max(t, N_0)$. The moved or added vertex is non-adjacent to x and to every current member of class x.*

Corollary 6. *For every $N_0 \le N \le N_1$, $\omega(G_N) \le M(N)$.*

Proof. Consider the state immediately after all vertices $\le N$ have been inserted. By (c), each class is an independent set, since classes only ever receive vertices non-adjacent to all current members, and removals preserve independence. By (b), every vertex $\le N$ lies in some class, and that class's anchor is $\le \max(N, N_0) = N$. Hence $V(G_N)$ is covered by the at most $M(N)$ classes with anchor $\le N$. □

The log is produced by `dyn_colour.c`. Each vertex is placed into a random compatible class; if none is found, one conflicting member is moved to another class (a depth-one augmenting step). Elements of P_{18} are pairwise adjacent and roughly as numerous as anchors, so at almost every time step exactly one class lacks a P_{18} element; the augmenting moves handle this. For $N_1 = 10^9$ the run inserts 105,158,686 vertices with 13,806,588 moves.

The log is checked by `verify_log.c`, which was written independently of the generator and uses different algorithms. It recomputes the vertex set using $\sqrt{-1} \equiv g^{q(q-1)/4} \pmod{q^2}$ (where g is a non-residue mod q), rather than Hensel lifting, and checks every root. It replays every operation and checks (a)–(c). It decides squarefreeness of every one of the 121,112,532 products $ab + 1$ claimed non-adjacent by trial division by all primes up to $(ab + 1)^{1/3}$, followed by an exact perfect-square test of the cofactor. If no prime $p \le n^{1/3}$ divides n, then n has at most two prime factors, so it is squarefree unless it is the square of a prime. The verifier rejects deliberately corrupted logs. All pairs were additionally re-checked with PARI/GP's `uissquarefree` (Section 6).

5. Large N: an explicit argument

Throughout, p, q denote primes, $\lambda(w) = \#\{u \bmod w : \ u^2 \equiv 1 \pmod{w}\}$, and $P_1 = P_2 = Q = 1000$.

Lemma 7 (large prime squares)**.** *For $1 \le w \le N$ and $V \ge 1$,*

$$\#\{z \le N : \ p^2 \mid wz + 1 \text{ for some } p > V\} \ \le \ \lambda(w)\Big(\frac{N^2+1}{V^2} + \frac{2(N+1)}{V} + 1\Big).$$

Proof. If $p^2 \mid wz + 1$ with $p > V$, write $wz + 1 = mp^2$. Then $1 \le m \le K := (wN + 1)/V^2$ and $mp^2 \equiv 1 \pmod{w}$. The pair (m, p) determines z. For fixed m, if u_0 satisfies $mu_0^2 \equiv 1 \pmod{w}$ then u_0 is a unit, and the solutions mod w are exactly $u_0 s$ with $s^2 \equiv 1$. So the integers $u \le U_m := \sqrt{(wN + 1)/m}$ with $mu^2 \equiv 1$ number at most $\lambda(w)(U_m/w + 1)$. Summing over $m \le K$ and using $\sum_{m \le K} m^{-1/2} \le 2\sqrt{K}$ gives at most $\lambda(w)\big(K + 2\sqrt{K(wN+1)}/w\big) = \lambda(w)\big(K + 2(wN + 1)/(wV)\big)$, which is at most the stated bound since $w \le N$. □

Writing $w = 2^e w'$ with w' odd, $\lambda(w) = c_e 2^{\omega(w')}$ with $c_e = 1, 1, 2, 4$ for $e = 0, 1, 2, \ge 3$. Hence $\lambda_{\max}(N) := \max_{w \le N} \lambda(w)$ is attained at 2^e times a product of initial odd primes, and is computed exactly.

Lemma 8 (one witness, P_7/P_{18} target)**.** *Let Z be a residue class modulo 100 contained in P_7 (or in P_{18}), and let $w \le N$ satisfy $25 \nmid wz+1$ and $4 \nmid wz+1$ for $z \in Z$. Let $\delta = 1 - \prod_{p\in\{3,7,11,13\}}(1-p^{-2})$. Then for every $V > P_1$,*

$$\#\{z \in Z \cap [1,N] :\ wz+1 \text{ non-squarefree}\} \le \tfrac{N}{100}\Big(\delta + \sum_{13<p\le P_1} p^{-2}\Big) + 15 + \pi(P_1) - \pi(13) + E_X,$$

where E_X is the number of z in the whole class $z \equiv c \pmod{25}$ for which $p^2 \mid wz+1$ for some $p > P_1$. Moreover

$$E_X \le \sum_{P_1<p\le V} \Big(\frac{N}{25p^2}+1\Big) + \lambda(w)\Big(\frac{N^2+1}{V^2} + \frac{2(N+1)}{V} + 1\Big).$$

Proof. The primes 2 and 5 are excluded by hypothesis. For $p \in S = \{3,7,11,13\}$ with $p \nmid w$, the condition $p^2 \mid wz+1$ is a single residue class mod p^2. By inclusion–exclusion over nonempty subsets $D \subseteq S$, each term counts a single class mod $100\prod_{p\in D} p^2$ in $[1,N]$, and so differs from its density by at most 1. This gives the first term, with error at most $2^4 - 1 = 15$; dropping primes dividing w only decreases the density. For $13 < p \le P_1$ use the union bound, each class contributing at most $N/(100p^2)+1$. For $p > P_1$, bound over the whole residue class mod 25: at most $N/(25p^2)+1$ for $P_1 < p \le V$, and Lemma 7 for $p > V$. □

The quantity E_X depends only on w and the class mod 25, so when a single witness bounds several cells of the same part, E_X is paid once. If $(w \bmod 4)(z \bmod 4) \equiv 3$ the cell is *free*: we only use the trivial bound $\lfloor N/100 \rfloor + 1$.

Lemma 9 (the large-q part of R)**.** *Let $L = \{n \le N : q^2 \mid n^2+1 \text{ for some } q > Q\}$. For every $V' > Q$,*

$$|L| \le T_R := \sum_{\substack{Q<q\le V'\\ q\equiv 1(4)}} \Big(2\Big\lfloor \frac{N}{q^2}\Big\rfloor + 2\Big) + \frac{N^2+1}{V'^2}\Big(\frac{1}{2} + \frac{\log(2N+1)}{2\log(1+\sqrt{2})}\Big).$$

Proof. For $q \le V'$, $S_q \cap [1,N]$ consists of two progressions. If $q > V'$ and $q^2 \mid n^2+1$, write $n^2+1 = kq^2$ with $k \le (N^2+1)/V'^2$. Here $k \ge 2$ is not a square, since $n^2+1 = (aq)^2$ is impossible for $n \ge 1$. Positive solutions of $x^2 - ky^2 = -1$ are units of norm -1 in $\mathbb{Z}[\sqrt{k}]$, so they are of the form ε^{2j+1}, where $\varepsilon \ge 1+\sqrt{2}$ is the fundamental unit of that order. Since $y\sqrt{k} = \sqrt{x^2+1} \le x+1$, we get $\varepsilon^{2j+1} \le 2x+1 \le 2N+1$. So for each k there are at most $\frac{1}{2} + \log(2N+1)/(2\log(1+\sqrt{2}))$ values of n. □

Definition 10. An element $h \in P_7 \cup P_{18}$ is *special* if $q^2 \mid h^2+1$ for some prime $q \ge 13$.

Lemma 11 (one witness, R target)**.** *Let $h \le N$ with $h \in P_7 \cup P_{18}$, let $\tau \in \mathbb{Z}/4$ with $(h \bmod 4)\tau \not\equiv 3$, and let $V > P_2 \ge Q$. The number of $n \in R \setminus L$ with $n \equiv \tau \pmod 4$ and $hn+1$ non-squarefree is at most*

$$\sum_{\substack{13\le q\le Q\\ q\equiv 1(4)}} 2\Big[\frac{N}{4q^2}\Big(\delta + \sum_{13<p\le P_2} p^{-2}\Big) + 15 + \pi(P_2) - \pi(13)\Big] + E_C + \mathrm{SP},$$

where $E_C \le \sum_{P_2<p\le V}(N/p^2+1) + \lambda(h)\big((N^2+1)/V^2 + 2(N+1)/V + 1\big)$. Here $\mathrm{SP} = 0$ if h is not special. In general $\mathrm{SP} \le \sum(N/(4q^2)+1)$, summed over the smallest primes $q \ge 13$, $q \equiv 1(4)$, whose squares have product at most $(N^2+1)/25$.

Proof. Every $n \in R \setminus L$ lies in one of the classes $\pm i_q \bmod q^2$ with $13 \le q \le Q$. Fix such a class. By Lemma 3, 5 and 2 cannot help. For $p \in \{3,7,11,13\}$ and $13 < p \le P_2$ with $p \ne q$, argue as in Lemma 8 inside the progression $n \equiv \pm i_q\ (q^2)$, $n \equiv \tau\ (4)$. The prime $p = q$ is special: on

the class, $hn+1$ is constant modulo q^2, and $q^2 \mid hn+1$ forces $n \equiv -h^{-1}$, which is compatible with $n^2 \equiv -1$ iff $h^2 \equiv -1 \pmod{q^2}$, i.e. iff h is special at q. In that case one whole class $n \equiv h \pmod{q^2}$ is compatible, contributing at most $N/(4q^2)+1$. The set of q at which h is special satisfies $25\prod q^2 \mid h^2+1 \le N^2+1$, so its contribution is at most that of the smallest admissible primes. Primes $p > P_2$ are handled globally, over all $n \le N$, by Lemma 7. □

5.1. **Patterns.** Let $A \subseteq [1,N]$ be a clique with $A \not\subseteq P_7$ and $A \not\subseteq P_{18}$; otherwise $|A| \le M(N)$ trivially. Split each of X and Y into four *cells* by residue mod 4. Record, for each cell, whether it is empty, nonempty with all elements special, or contains a non-special element. For C, record which residues mod 4 occur. This gives $3^4 \cdot 3^4 \cdot 2^4$ possible *patterns*.

Given the pattern, we bound each part using at most two witness residues from the other parts. We use any element of $Y \cup C$ for X and of $X \cup C$ for Y. For C we use non-special elements of $X \cup Y$, or special ones at the cost of SP. Each cell is bounded by the minimum of the applicable bounds:

- the trivial bound;
- Lemma 8 or 11 for the chosen witness;
- for a special-only cell of X or Y, the bound $\sum_{13 \le q \le Q} 2(\lfloor N/(100q^2) \rfloor + 1)$, since such elements are either in some S_q with $q \le Q$ or lie in L.

Each witness's E_X or E_C is added once per part. Finally we add T_R once, for $A \cap L$. Elements of L may be counted twice; this only weakens the bound. Let $\mathrm{UB}(N)$ be the maximum of the resulting quantity over all patterns. Then $|A| \le \mathrm{UB}(N)$.

Proposition 12. *There are* $5 \cdot 10^8 = N^{(0)} < N^{(1)} < \cdots < N^{(k)} = 10^{35}$ *with* $\mathrm{UB}(N^{(i+1)}) < M(N^{(i)})+1$ *for all* i. *The maximum of* $\mathrm{UB}(N^{(i+1)})/(M(N^{(i)})+1)$ *is* 0.9960, *attained on the first interval.*

For $N^{(i)} \le N \le N^{(i+1)}$, a clique in G_N is also a clique in $G_{N^{(i+1)}}$, so $|A| \le \mathrm{UB}(N^{(i+1)}) < M(N^{(i)})+1 \le M(N)+1$. Hence Theorem 1 holds on $[5 \cdot 10^8, 10^{35}]$.

Proposition 13 (tail)**.** *For* $N \ge 10^{35}$, $\mathrm{UB}(N) < M(N)+1$.

Sketch. Take $V = V' = N^{2/3}$. Every term of $\mathrm{UB}(N)$ is either linear in N or sublinear. The linear part is at most $U_{\mathrm{lin}}N$, where U_{lin} is the pattern maximum of the linear coefficients; infinite prime sums are bounded by exact partial sums up to $3 \cdot 10^8$ plus $\sum_{n>3\cdot 10^8,\, n \text{ odd}} n^{-2}$. The sublinear part $S(N)$ comes from at most six witness applications, T_R, and the bounded inclusion–exclusion and rounding constants. It is bounded using $\pi(x) < 1.25506\, x/\log x$ [5] and $\lambda(w) \le 4 \cdot 2^{\omega(w)} \le 4 \cdot 2^{1.38402 \log N/\log\log N}$ [6]. Each resulting term of $S(N)/N$ is non-increasing for $N \ge 10^8$. Since $M(N)+1 > N/25$, it suffices that $25(U_{\mathrm{lin}} + S(10^{35})/10^{35}) < 1$. In interval arithmetic this quantity is at most 0.7484, with $25U_{\mathrm{lin}} \le 0.6907$. □

The computations behind Propositions 12 and 13 are in `analytic_rig.py`. All real quantities are evaluated as Arb balls [7], and each is replaced by the exact rational upper endpoint of its ball; prime counts and floors are exact integers. The maximisation over patterns is then carried out in exact rational arithmetic. The limiting value $\mathrm{UB}(N)/M(N) \to 0.691$ comes from cliques contained in R.

6. Verification and reproducibility

- **Small N:** `maxclique_small 10000` and `bk_check.py 5000`: no mismatches.
- **Certificate:** `dyn_colour 1000000000 9982 1 log.bin` (about one hour) and `verify_log 1000000000 9982 log.bin`: VERIFIED, 121,112,532 pairs, 0 failures.

- **PARI/GP cross-checks:** PARI's `uissquarefree` confirms that all 121,112,532 certificate products $ab + 1$ are squarefree. Recomputing the vertex set up to 10^9 as $\{n : \texttt{uissquarefree}(n^2 + 1) = 0\}$ gives 105,158,686 vertices, with the same sum and the same sum of squares (mod 2^{64}) as the verifier's vertex set.
- **Analytic range:** `analytic_rig.py grid 5e8 1e35` reports 181 intervals, all satisfied, with worst ratio 0.996034. `analytic_rig.py tail 1e35` reports $0.748362 < 1$. The grid's top endpoint and the tail's starting point are the same integer, $\lfloor \mathrm{float}(10^{35}) \rfloor \approx 10^{35}$.
- **Independent review:** an independent AI re-implementation of Section 5, written without access to our code, found $\mathrm{UB}/M \leq 0.763$ on $[5 \cdot 10^8, 10^{13}]$. Its lemma spot-checks against brute force found no violations.

Acknowledgements. We thank M. Sawhney, W. van Doorn, D. Weisenberg, B. Alexeev, S. Cambie and N. Sothanaphan, whose results and comments on the problem's forum thread this work builds on, and T. Bloom for maintaining `erdosproblems.com`.

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